

Myles Allred

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EDUCATION

Aug 2022 –
Expected May 2026

University of Illinois – Chicago, Illinois

BS, Neuroscience

BS, Mathematical Computer Science (concentration in computational mathematics)

Cumulative GPA: 4.0

Dean's list all semesters

June 2022

Eureka Highschool, Missouri

RESEARCH

Feb 2024 – Present

University of Illinois Chicago - Center for Alcohol Research in Epigenetics

With Dr. Katherine McMurray:

Investigating neurobiology underlying comorbid psychiatric disorders at behavioral, molecular, and circuit level.

Characterized a novel rodent model of comorbid alcohol use disorder/panic disorder using homeostatic-stressor CO₂-inhalation as panicogenic challenge, finding that withdrawal from chronic ethanol increased sensitivity to CO₂-inhalation in C57Bl/6J but not DBA/2J mice. Developed custom FIJI script enabling user-friendly, Ilastik powered quantification of custom regions of interest for cFos positive cells in images (see Github). Manuscript in preparation. Posters: UIC Dept. Of Psychiatry, SfN 2024, RSA 2025.

Investigating role of microglial acid sensor TDAG8 in regulating AUD-associated outcomes and drinking behaviors. Studied behavioral, molecular, and physiological changes. Poster: UIC Undergraduate Research Forum.

Currently investigating brain-wide functional network disruptions in model of comorbid AUD/PD, using tissue-clearing techniques to quantify cFos expression. Developing network analysis to understand functional changes underlying differential sensitivity to CO₂ and identify hub regions modulating these changes. My honors thesis is on this project; will result in another publication.

May 2025 – Aug 2025

University of Illinois Chicago - Dept. of Math, Stats, Comp. Sci.

With Jennifer Vaccaro (reading project):

Statistical mechanics and neural systems: Ising models, message passing, Metropolis Monte Carlo simulations, replica method, replica symmetry breaking, dimensionality reduction. Developed toolkit in python (see Github), enabling simulation and analysis of Ising systems.

Talk: From Monte Carlo Simulations to Analytic Solutions of Ising Models

Jan 2025 – May 2025 *University of Illinois Chicago - Dept. of Math, Stats, Comp. Sci.*
With Jennifer Vaccaro (reading project):
 Graph representation learning; graph convolutional networks and their relation to image convolution and spectral theory; graph theoretic measures relevant to neural networks.
Talk: Graph Convolutional Networks (as an extension of convolution over grids)

AWARDS

May 2025 UIC Honor College – *Conference Travel Grant*
 May 2025 Research Society on Alcohol – *Student Merit and Junior Investigator Meeting Grant*
 April 2025 UIC Laboratory of Integrative Neuroscience – *Featherstone Scholarship*
 April 2025 UIC Laboratory of Integrative Neuroscience – *Founders Award in Systems & Behavioral Neuroscience*
 April 2024 UIC French Dept. – *Student Showcase Award*

MEMBERSHIP

April 2025 - Present Nu Rho Psi – University of Illinois Chicago Chapter
 April 2025 - Present Phi Beta Kappa – University of Illinois Chicago Chapter
 2025 - Present Research Society on Alcohol
 2024 - Present Society for Neuroscience
 2024 - Present UIC Quantum Information Science Society (club officer June 2025-present)
 Oct 2022 – May 2023 UIC Engineers for a Sustainable World

OTHER

EXPERIENCE

June 2025 – Aug 2025 *UIC Math, Stats, Comp. Sci. Dept. – Young Scholars Program*
 Coordinated by Dr. Brooke Shipley, Dr. Vishesh Jain, Nick Christo, Lexie Wheeler.
 Undergraduate teaching instructor at math program for Chicago high school students.
 Worked with graduate students to introduce wide range of topics in college math.
 Helped with lectures and mentored students in small groups.

Jan 2024 – May 2024 *UIC Biological Sciences Department – BIOS 102*
 Peer mentor/teaching assistant for incoming biology students.

May 2023 – Aug 2023 *Civil Design Inc. – Land Survey Intern*
 Conducted land, construction, and legal surveys. Created, analyzed, and presented on datasets to increase team efficiency in survey and engineering projects.

SKILLS

Programming: Python (Numpy, Pandas, PyTorch, Matplotlib), SageMath, Java
Animal husbandry (mice): IP Injections, fear conditioning, plethysmography, CO₂ exposure model, colony management, blood draws, transcatheter perfusion
Wet lab: Immunohistochemistry, brightfield and fluorescent imaging, qPCR, iDISCO
Languages: English (native), French (B1)

PRESENTATIONS

November 2025	<u>UIC College of Medicine Research Forum – poster:</u> “A Novel Mouse Model Relevant to AUD-Panic Disorder Comorbidity: Strain Differences in Sensitivity to CO2 Inhalation During Withdrawal from Chronic Ethanol”
October 2025	<u>UIC Dept. of Psychiatry 16th Annual Research Forum – poster:</u> “A Novel Mouse Model Relevant to AUD-Panic Disorder Comorbidity: Strain Differences in Sensitivity to CO2 Inhalation During Withdrawal from Chronic Ethanol”
August 2025	<u>UIC Directed Reading Program – talk:</u> “From Monte Carlo Simulations to Analytic Solutions of Ising Models”
June 2025	<u>Research Society on Alcohol meeting – poster:</u> “A Novel Mouse Model Relevant to AUD-Panic Disorder Comorbidity: Strain Differences in Sensitivity to CO2 Inhalation During Withdrawal from Chronic Ethanol”
May 2025	<u>UIC Directed Reading Program – talk:</u> “Graph Convolutional Networks (as an extension of convolution on grids)”
November 2024	<u>UIC Honor College Undergraduate Research Symposium – poster:</u> “Microglial acid sensing regulates AUD associated outcomes and drinking behavior”
October 2024	<u>Society for Neuroscience meeting – poster:</u> “Withdrawal from chronic alcohol increases CO2 inhalation sensitivity in mice and rats: a novel model relevant to panic and PTSD comorbidity following AUD”
September 2024	<u>UIC Dept. of Psychiatry 15th Annual Research Forum – poster:</u> “Developing a novel model relevant to AUD-panic disorder comorbidity: Strain differences in sensitivity to CO2 inhalation during withdrawal from chronic alcohol”